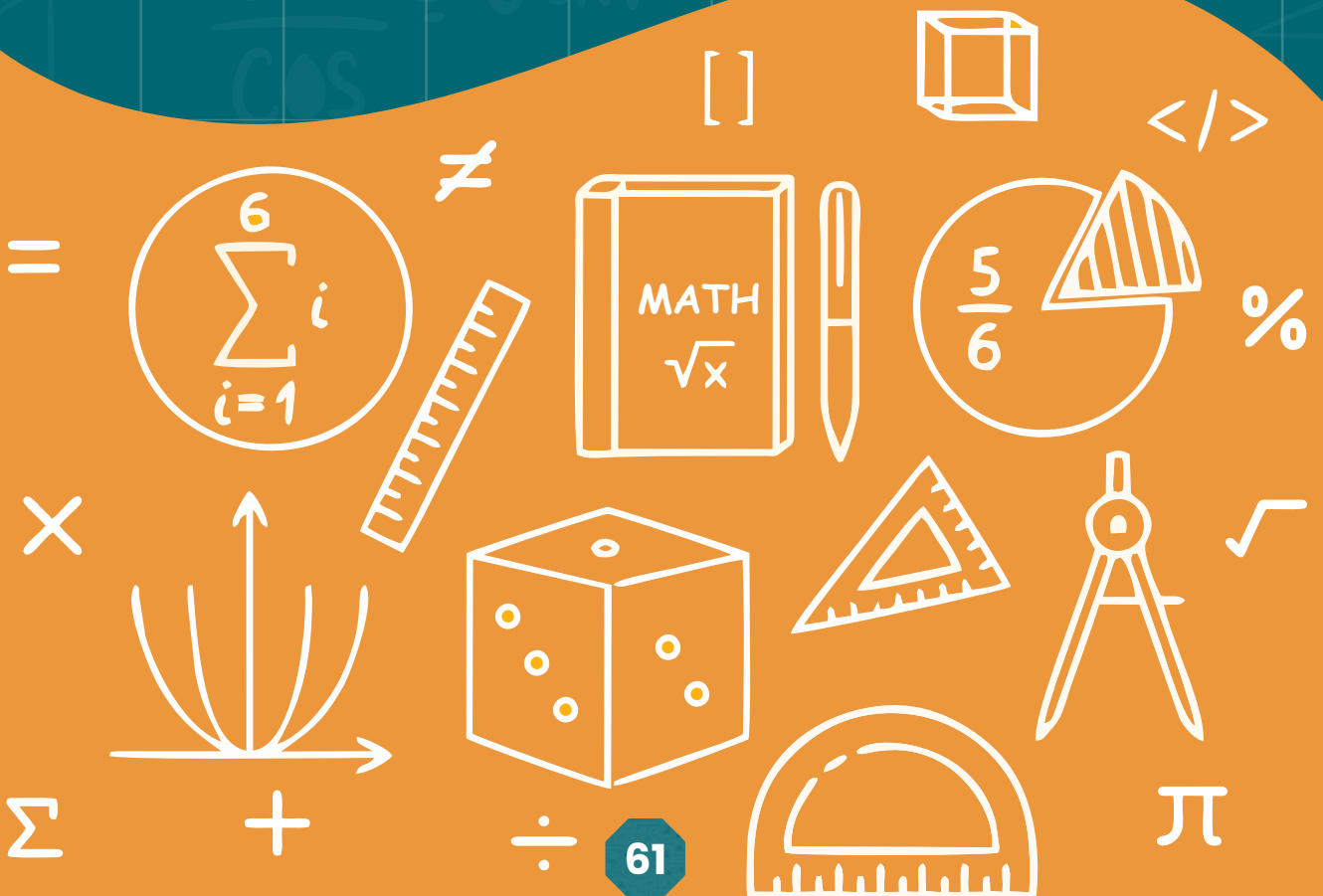


UNIT 6

FACTORIZATION, SIMULTANEOUS EQUATIONS



BASIC ALGEBRAIC FORMULAS

Following are the three widely used algebraic formulae:

$$(i) (a + b)^2 = a^2 + 2ab + b^2 \quad (ii) (a - b)^2 = a^2 - 2ab + b^2$$

$$(iii) a^2 - b^2 = (a + b)(a - b)$$

You can verify them by performing multiplication.

$$(a + b)^2 = (a + b)(a + b) \\ = a^2 + ab + ab + b^2 = a^2 + 2ab + b^2$$

$$(a - b)^2 = (a - b)(a - b) \\ = a^2 - ab - ab + b^2 = a^2 - 2ab + b^2$$

$$(a + b)(a - b) = a^2 - ab + ab - b^2 \\ = a^2 - b^2$$

Let us apply them to solve some problems.

Example 1. Evaluate:

$$(i) (102)^2$$

$$(ii) (98)^2$$

$$(iii) (102)(98)$$

Solution:

$$(i) (102)^2 = (100 + 2)^2 \\ = (100)^2 + 2(100)(2) + (2)^2$$

$$[Using: (a + b)^2 = a^2 + 2ab + b^2] \\ = 10000 + 400 + 4 = 10404$$

$$(iii) (102)(98) = (100 + 2)(100 - 2) \\ = (100)^2 - (2)^2$$

$$[Using: (a + b)(a - b) = a^2 - b^2] \\ = 10000 - 4 = 9996$$

$$(ii) (98)^2 = (100 - 2)^2 \\ = (100)^2 - 2(100)(2) + (2)^2$$

$$[Using: (a - b)^2 = a^2 - 2ab + b^2] \\ = 10000 - 400 + 4 = 9604$$

Activity: Solve 203×197

$$(200 + \underline{\hspace{2cm}}) \times (\underline{\hspace{2cm}} - 3)$$

$$Using (a + b)(a - b) = (\underline{\hspace{2cm}})$$

$$= (200)^2 - (\underline{\hspace{2cm}})^2$$

$$= 40,000 - \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$$

Example 2. Find the value $x^2 + \frac{1}{x^2}$ and $x^4 + \frac{1}{x^4}$ when (i) $x + \frac{1}{x} = 6$

$$(ii) x - \frac{1}{x} = 2$$

Solution:

$$(i) x + \frac{1}{x} = 6 \dots \text{Given} \quad (\text{Squaring both sides})$$

$$\left(x + \frac{1}{x}\right)^2 = (6)^2$$

$$x^2 + 2\left(x\right)\left(\frac{1}{x}\right) + \frac{1}{x^2} = 36$$

$$x^2 + 2 + \frac{1}{x^2} = 36$$

$$x^2 + \frac{1}{x^2} = 36 - 2$$

$$x^2 + \frac{1}{x^2} = 34$$

$$\left(x^2 + \frac{1}{x^2}\right)^2 = (34)^2 \quad (\text{Squaring both sides})$$

$$x^4 + 2\left(x^2\right)\left(\frac{1}{x^2}\right) + \frac{1}{x^4} = 1156$$

$$x^4 + \frac{1}{x^4} = 1156 - 2 = 1154$$

(ii) $x - \frac{1}{x} = 2$... **Given** (Squaring both sides)

$$\left(x - \frac{1}{x}\right)^2 = (2)^2$$

$$x^2 - 2\left(x\right)\left(\frac{1}{x}\right) + \frac{1}{x^2} = 4$$

$$x^2 - 2 + \frac{1}{x^2} = 4$$

$$x^2 + \frac{1}{x^2} = 4 + 2$$

$$x^2 + \frac{1}{x^2} = 6$$

$$\left(x^2 + \frac{1}{x^2}\right)^2 = (6)^2 \quad (\text{Squaring both sides})$$

$$x^4 + 2\left(x^2\right)\left(\frac{1}{x^2}\right) + \frac{1}{x^4} = 36$$

$$x^4 + \frac{1}{x^4} = 36 - 2 = 34$$

Example 3. Show that: $x^4 + \frac{1}{x^4} = \left(x^2 + \frac{1}{x^2}\right)^2 - 2$

Proof:

$$\text{R.H.S} = \left(x^2 + \frac{1}{x^2}\right)^2 - 2$$

$$= (x^2)^2 + 2\left(x^2\right)\left(\frac{1}{x^2}\right) + \left(\frac{1}{x^2}\right)^2 - 2$$

$$= x^4 + 2 + \frac{1}{x^4} - 2 = x^4 + \frac{1}{x^4} = \text{L.H.S. Hence proved.}$$

EXERCISE 6.1

Q1. Evaluate the following by using suitable formula:

- | | |
|--------------|--------------------|
| 1. $(105)^2$ | 5. 104×96 |
| 2. $(96)^2$ | 6. 47×53 |
| 3. $(57)^2$ | 7. 107×93 |
| 4. $(52)^2$ | 8. 89×111 |

Q2. Evaluate the following by using suitable formula:

- | | |
|---------------|-----------------------|
| 1. $(1.03)^2$ | 4. $(0.91)^2$ |
| 2. $(0.99)^2$ | 5. 1.03×0.97 |
| 3. $(1.05)^2$ | 6. 5.02×4.98 |

Q3. Find the value of $x^2 + \frac{1}{x^2}$ and $x^4 + \frac{1}{x^4}$ when:

- | | | |
|--------------------------|---------------------------|------------------------------------|
| 1. $x + \frac{1}{x} = 5$ | 2. $x - \frac{1}{x} = -4$ | 3. $\frac{1}{x} + \frac{1}{x} = 7$ |
|--------------------------|---------------------------|------------------------------------|

Q4. Find the value of $x^2 + \frac{1}{x^2}$ and $x^4 + \frac{1}{x^4}$ when:

- | | | |
|--------------------------|----------------------------|-------------------------------------|
| 1. $x - \frac{1}{x} = 3$ | 2. $x - \frac{1}{x} = 0.2$ | 3. $\frac{1}{x} - \frac{1}{x} = -6$ |
|--------------------------|----------------------------|-------------------------------------|

Q5. Show that:

- | | |
|---|---|
| 1. $x^2 + \frac{1}{x^2} = \left(x + \frac{1}{x}\right)^2 - 2$ | 2. $x^2 + \frac{1}{x^2} = \left(\frac{1}{x} - x\right)^2 + 2$ |
|---|---|

FACTORIZATION

If the product of two or more expressions is equal to the given expression, then these two or more expressions are called the factors of the given expression.

For example: if $3x^2 + 6x = 3x(x+2)$ then, 3, x and $(x+2)$ are the factors of $3x^2 + 6x$.

The process of finding factors of a given expression is called factorization. We express the given expression as a product of two or more expression.

Factors of the expression of the type: $ka + kb + kc$

We know that if k is a non-zero real number then

$$ka + kb + kc = k(a + b + c)$$

Here k and $(a + b + c)$ are the factors of $k(a + b + c)$

Factors of the expression of the type: $ac + ad + bc + bd$

We show that: $ac + ad + bc + bd = (a + b)(c + d)$

Proof:

$$\begin{aligned} \text{L.H.S.} &= ac + ad + bc + bd = a(c + d) + b(c + d) \\ &= (a + b)(c + d) = \text{R.H.S.} \end{aligned}$$

Example 2. Factorize:

(i) $5x + xz + 5z + z^2$

(ii) $3x^2y + 6xy^2 - 2xz - 4yz$

Solution:

$$\begin{aligned} \text{(i)} \quad 5x + xz + 5z + z^2 \\ &= x(5 + z) + z(5 + z) \\ &= (5 + z)(x + z) \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad 3x^2y + 6xy^2 - 2xz - 4yz \\ &= 3xy(x + 2y) - 2z(x + 2y) \\ &= (x + 2y)(3xy - 2z) \end{aligned}$$

Factors of the expression of the type: $a^2 + 2ab + b^2$ and $a^2 - 2ab + b^2$

(i) We show that:

$$a^2 + 2ab + b^2 = (a + b)^2$$

Proof:

$$\begin{aligned} \text{L.H.S.} &= a^2 + 2ab + b^2 \\ &= a^2 + ab + ab + b^2 \\ &= a(a + b) + b(a + b) \\ &= (a + b)(a + b) \\ &= (a + b)^2 = \text{R.H.S.} \end{aligned}$$

(ii) We also show that:

$$a^2 - 2ab + b^2 = (a - b)^2$$

Proof:

$$\begin{aligned} \text{L.H.S.} &= a^2 - 2ab + b^2 \\ &= a^2 - ab - ab + b^2 \\ &= a(a - b) - b(a - b) \\ &= (a - b)(a - b) = (a - b)^2 = \text{R.H.S.} \end{aligned}$$

Example 1. Factorize: $x^2 + 6xy + 9y^2$

Solution:

$$\begin{aligned} x^2 + 6xy + 9y^2 \\ &= (x)^2 + 2(x)(3y) + (3y)^2 \\ &= (x + 3y)^2 \end{aligned}$$

Example 2. Factorize: $25x^2 - 10xy + y^2$

Solution:

$$\begin{aligned} 25x^2 - 10xy + y^2 \\ &= (5x)^2 - 2(5x)(y) + (y)^2 = (5x - y)^2. \end{aligned}$$

Factors of the expression of the type: $a^2 - b^2$

We show that:

$$a^2 - b^2 = (a + b)(a - b)$$

Proof: $\text{R.H.S.} = (a + b)(a - b)$

$$\begin{aligned} &= a(a - b) + b(a - b) \\ &= a^2 - ab + ab - b^2 \\ &= a^2 - b^2 = \text{L.H.S.} \end{aligned}$$

Example 1. Factorize: $4a^2 - 9b^2$

Solution: $4a^2 - 9b^2$
 $= (2a)^2 - (3b)^2$
 $= (2a + 3b)(2a - 3b)$

6.2.5 Factors of the expression of the type: $a^2 \pm 2ab + b^2 - c^2$

In this type of expressions, we first apply the formula:

$$x^2 + 2xy + y^2 = (x + y)^2 \text{ and then } a^2 - b^2 = (a + b)(a - b)$$

Example 1. Factorize $x^2 + 2(x)(5) + 25 - y^2$

Solution:

$$x^2 + 10x + 25 - y^2 = [(x)^2 + (x)(10) + (5)^2] - y^2$$

$$= (x + 5)^2 - y^2 = (x + 5 + y)(x + 5 - y)$$

Similarly, we factorize the expression of the type $a^2 - 2ab + b^2 - c^2$ by using the formulas $a^2 - 2ab + b^2 = (a - b)^2$ and $x^2 - y^2 = (x + y)(x - y)$

Example 2. Find factors of $x^2 - 2xy + y^2 - z^2$

Solution:

$$x^2 - 2xy + y^2 - z^2 = [(x)^2 - 2(x)(y) + (y)^2] - z^2$$

$$= (x - y)^2 - z^2 = (x - y + z)(x - y - z)$$

EXERCISE 6.2

A. Factorize the following:

(1) $4x + 8z$

(2) $5x + 10y + 30z$

(3) $2x - 4xy + 8xz$

(4) $2a^2 + 10a^3 - 20a^4$

(5) $3a^2b + 7ab^2 - 8a^2b^2$

(6) $6a^2bc + 12abc^2 - 36abc^2$

(7) $5x + 10y + 3xz + 6yz$

(8) $abc - abd + cx - xd$

(9) $x^2 + 5x + 6xy + 30y$

(10) $7xy + 14yz - 5ax - 10az$

(11) $3x^2 + 6y^2 + 6x^2z + 12y^2z$

(12) $x^2 - 7xy - xz + 7yz$

B. Resolve into factors:

(1) $a^2 + 10a + 25$

(2) $x^2 + 12xy + 36y^2$

(3) $4x^2 + 12xy + 9y^2$

(4) $16a^2 + 40ab + 25b^2$

(5) $b^2 + c^2 - 2bc$

(6) $49p^2 - 14p + 1$

(7) $81c^2 - 36cd + 4d^2$

(8) $144x^2 - 72xy + 9y^2$

(9) $3a - 6ab + 3b^2$

C. Find factors of the following:

(1) $81x^2 - 4y^2$

(2) $169x^2 - 100y^2$

(3) $3a^2 - 27b^2$

(4) $32x^2 - 18y^2$

(5) $5x^2 - 125y^2$

(6) $(a - b)^2$

(7) $(2a + b)^2 + 9c^2$

(8) $(x + y)^2 - z^2$

(9) $c^2 - (a - b)^2$

(10) $(x + y)^2$

(11) $(a + b)^2 - (p + q)^2$

(12) $144x^2 - y^2$

(13) $32x^2 - 50y$

D. Resolve into factors:

(1) $2x^2 + y^2 + z^2$

(2) $a^2 + b^2 - 2ab - 2c^2$

(3) $64a^2 + 48ab + 9b^2 - c^2$

(4) $4p^2 - 12pq + 9q^2 - 49r^2$

(5) $x^2 + y^2 - 252$

(6) $x^2 + 2xy - y^2 - z^2$

(7) $81c^2 + 16p^2 + 18cd + d^2$

(8) $36 - 25a^2 + 70ab - 49b^2$

6.3 MANIPULATION OF ALGEBRAIC EXPRESSION

Now we recognize and apply following formulas to solve problems.

6.3.1 Formula for the cube of the sum of two terms

$$(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

Proof: LHS = $(a + b)^3 = (a + b)(a + b)^2 = (a + b)(a^2 + 2ab + b^2)$

$$= a(a^2 + 2ab + b^2) + b(a^2 + 2ab + b^2)$$

$$= a^3 + 2a^2b + ab^2 + ba^2 + 2ab^2 + b^3$$

$$= a^3 + 3a^2b + 3ab^2 + b^3 = \text{RHS.}$$

Note: $(a + b)^3 = a^3 + b^3 + 3ab(a + b)$
 $a^3 + b^3 = (a + b)^3 - 3ab(a + b)$

Example 1. Find the value of $x^3 + \frac{1}{x^3}$ when $x + \frac{1}{x} = 5$

Solution:

(i) $x + \frac{1}{x} = 5$

Method I: From known to unknown.

Cubing both sides, by formula.

$$\left(x + \frac{1}{x}\right)^3 = (5)^3$$

$$x^3 + 3(x^2)\left(\frac{1}{x}\right) + 3(x)\left(\frac{1}{x^2}\right) + \frac{1}{x^3} = 125$$

$$x^3 + 3x + \frac{3}{x} + \frac{1}{x^3} = 125$$

$$x^3 + 3\left(x + \frac{1}{x}\right) + \frac{1}{x^3} = 125$$

$$\because x + \frac{1}{x} = 5$$

$$x^3 + 3(5) + \frac{1}{x^3} = 125$$

$$x^3 + \frac{1}{x^3} = 125 - 15 = 110.$$

Method II: From unknown to known.

We know that:

$$a^3 + b^3 = (a + b)^3 - 3ab(a + b)$$

$$x^3 + \frac{1}{x^3} = \left(x + \frac{1}{x}\right)^3 - 3\left(x\right)\left(\frac{1}{x}\right)\left(x + \frac{1}{x}\right)$$

$$x^3 + \frac{1}{x^3} = \left(x + \frac{1}{x}\right)^3 - 3\left(x + \frac{1}{x}\right)$$

$$= 5^3 - 3(5)$$

$$= 125 - 15$$

$$= 110.$$

6.3.2 Formula for the cube of the difference of two terms

$$(a - b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$$

Proof: L.H.S. = $(a - b)^3 = (a - b)(a - b)^2 = (a - b)(a^2 - 2ab + b^2)$

$$= a(a^2 - 2ab + b^2) - b(a^2 - 2ab + b^2)$$

$$= a^3 - 2a^2b + ab^2 - a^2b + 2ab^2 - b^3$$

$$= a^3 - 3a^2b + 3ab^2 - b^3 = \text{R.H.S.}$$

EXERCISE 6.3

Q1. find the cube of each of the following:

1. $2x + 3$

5. $x + 7z$

8. $5 - \frac{a}{4}$

2. $5x - 1$

6. $\frac{x}{2} - 1$

9. $5a - 3b$

3. $2x - 3y$

7. $5 + 3a$

10. $\frac{x}{3} - \frac{y}{5}$

4. $2a + 5b$

Q2. Find the cube of each of the following by using the formula:

1. 18

2. 13

3. 105

4. 10.1

5. 0.98

SIMULTANEOUS LINEAR EQUATIONS

We know that a linear equation is an equation with degree one. It may be in one, two or more variables.

For example, $2x + 6 = 0$ is a linear equation in one variable x

Similarly $5x + 6y - 9 = 0$ is a linear equation in two variables x and y .

Standard form of a linear equation in one variable is: $ax + b = 0$,

where a, b are real numbers and $a \neq 0$. For example, $2x + 3 = 0$.

Standard form of a linear equation in two variables is: $ax + by + c = 0$, where a, b, c are real numbers with $a \neq 0$ and $b \neq 0$.

Recognize Simultaneous Linear Equations in one and two variables

Two or more linear equations which occur in one and the same problem

are called simultaneous linear equations.

Example 1. $2x + 3 = 5$ and $2x + y = 6$ are two simultaneous linear equations in one and two variables respectively.

Example 2. $6x + y = 8$ and $3x - y = 1$ are two simultaneous linear equations in two variables.

Note: Simultaneous linear equations have only one solution. As in example 1, the solution is $x = 1, y = 4$ or $(x, y) = (1, 4)$. In example 2 the solution is $x = 1, y = 2$ or $(x, y) = (1, 2)$.

Give the concept of formation of a linear equation in two variables.

We know that a linear equation in two variables is in fact an algebraic representation of a problem or statement involving two quantities. In order to form a linear equation we have to follow following steps.

Step 1. Read the problem or statement carefully and identify the unknown quantities ($x, y, z, t, \dots$)

Step 2. Represent the unknown quantities with letters of the English alphabet.

Step 3. Write the equation which satisfies the condition of given problem or statement.

Example 1. Form a linear equation for each of the following statements.

- (i) The sum of ages of two students is 30 years.
- (ii) Total amount spent for 5 kilograms of apples and 3 kilograms of mangoes is 550 rupees.
- (iii) Price of 6 pens is double the price of a book.

Solution:

(i) Let the ages of two students in years be x and y . So, the required equation is: $x + y = 30$.

(ii) Let the price of 1 kg of apples be x rupees and the price of 1kg of mangoes be y rupees. The required equation will be $5x + 3y = 550$

(iii) Let the price of a pen be x rupees and that of a book be y rupees. So, the required equation will be $6x = 2y$ or $6x - 2y = 0$

Example 2. Form two simultaneous linear equations in two variables for the statement. The length of a rectangle is greater than its width by 4 cm and perimeter of the rectangle is 192 cm.

Solution: Let length= y and width = x

Then $y - x = 4$... (i)

Now Perimeter = $2(x + y) = 192$ cm

or $x + y = 96$... (ii)

Important facts about linear equations:

(1) A single linear equation in two unknowns is satisfied by as many pair of values as required.

For example, $x + y = 6$ is a single linear equation in two variables and its solutions are infinite. Some of them are (1, 5), (2, 4), (3, 3), (4, 2), etc.

Verification: We have $x + y = 6$,

for (1, 5), we get $1 + 5 = 6$ (satisfied) for (2, 4), we get $2 + 4 = 6$ (satisfied)

for (3, 3), we get $3 + 3 = 6$ (satisfied) for (4, 2), we get $4 + 2 = 6$ (satisfied).

(2) Two simultaneous linear equations in two unknowns have only one solution (i.e., one pair of values).

Either both of these are in two variables or one of them is in one variable.

Note: Solution of two simultaneous equations means the values of the unknowns which satisfy both the equations.

For example, $x + y = 5$ and $x - y = 3$ are two simultaneous linear equations and they have only one solution, i.e., (4, 1).

Verification: We have $x + y = 5$ and $x - y = 3$,

for (4, 1), we get $4 + 1 = 5$ and $4 - 1 = 3$

So both equations are satisfied by (4, 1).

EXERCISE 6.4**1. Identify and separate linear equations in one variable, linear equations in two variables, and simultaneous linear equations.**

(i) $4x + 5y = 12$

(ii) $3x - 9 = 0$

(iii) $5x = 10, x + 2y = 15$

(iv) $y - x = 4, x + y = 20$

(v) $7y = 14$

(vi) $2y - 6 = 5x$

2. Form a linear equation for each of the following statements:

(i) The sum of the ages of two brothers is 40 years.

(ii) The cost of five books is twice the cost of a notebook.

(iii) If three times a number is subtracted from 10, the result is 16.

(iv) If the cost of 4 kg of apples is added to the cost of 2 kg of oranges, the total is 90 rupees.

(v) The number of students in a class is four-fifths of the number of chairs in the classroom.